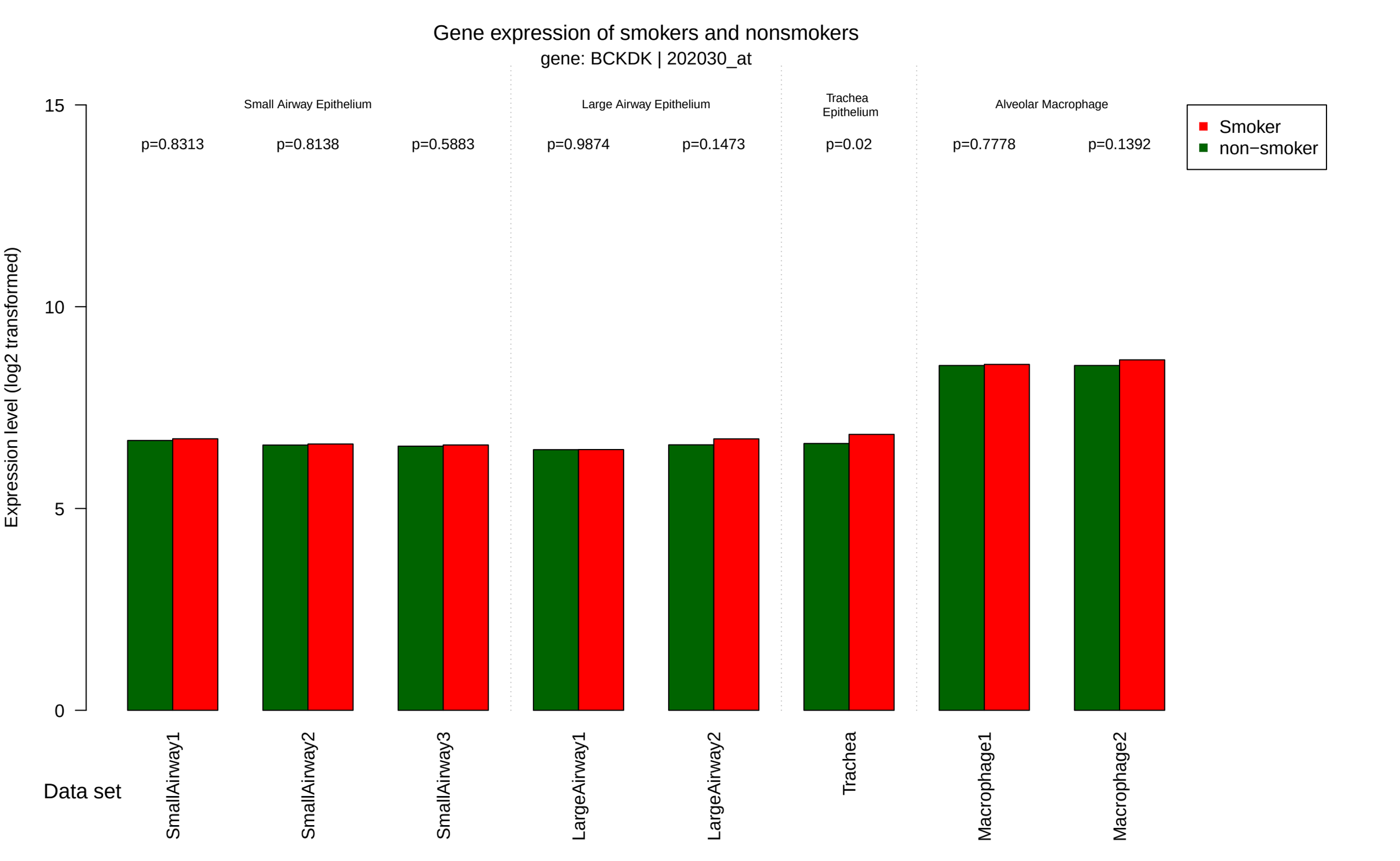
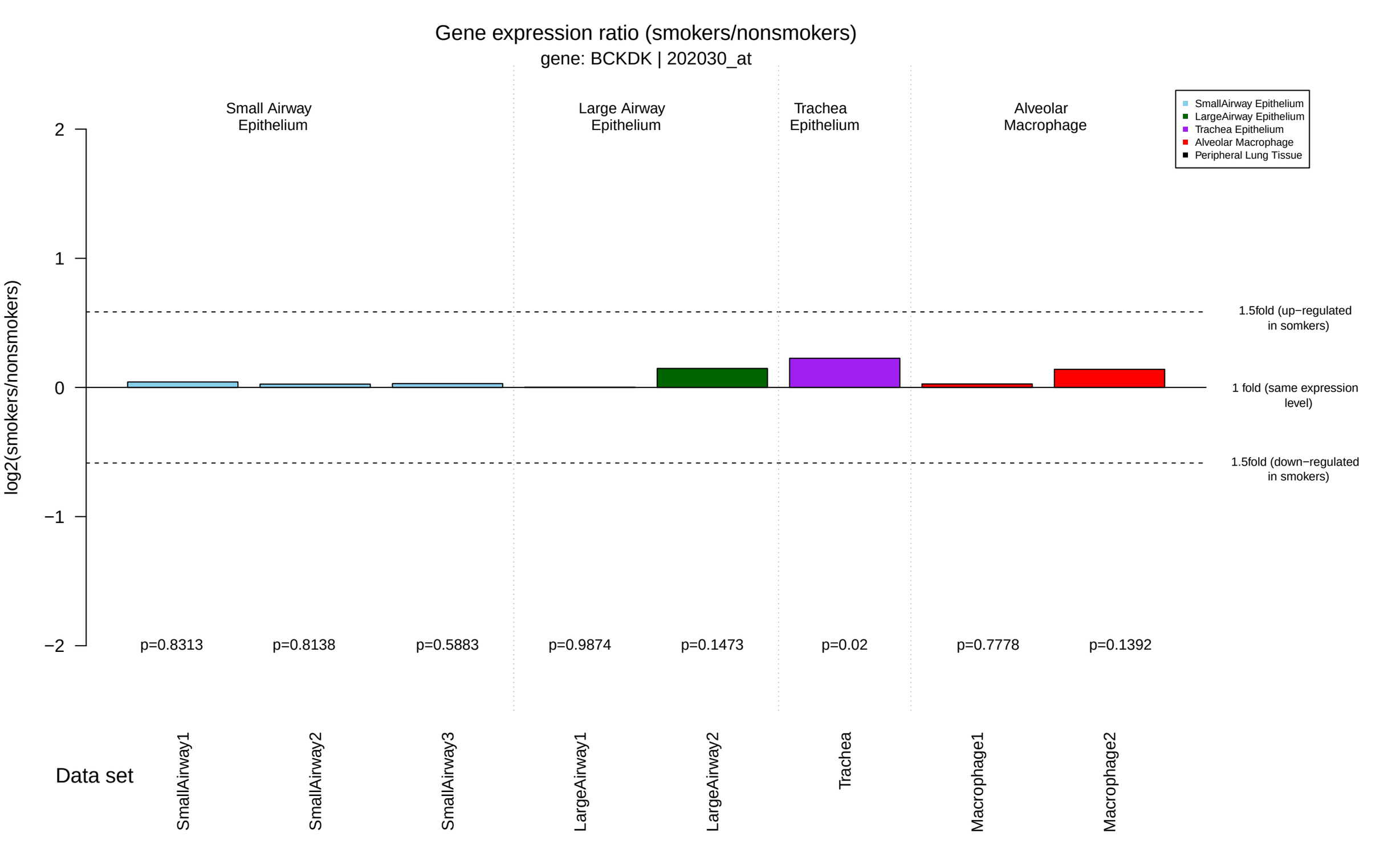
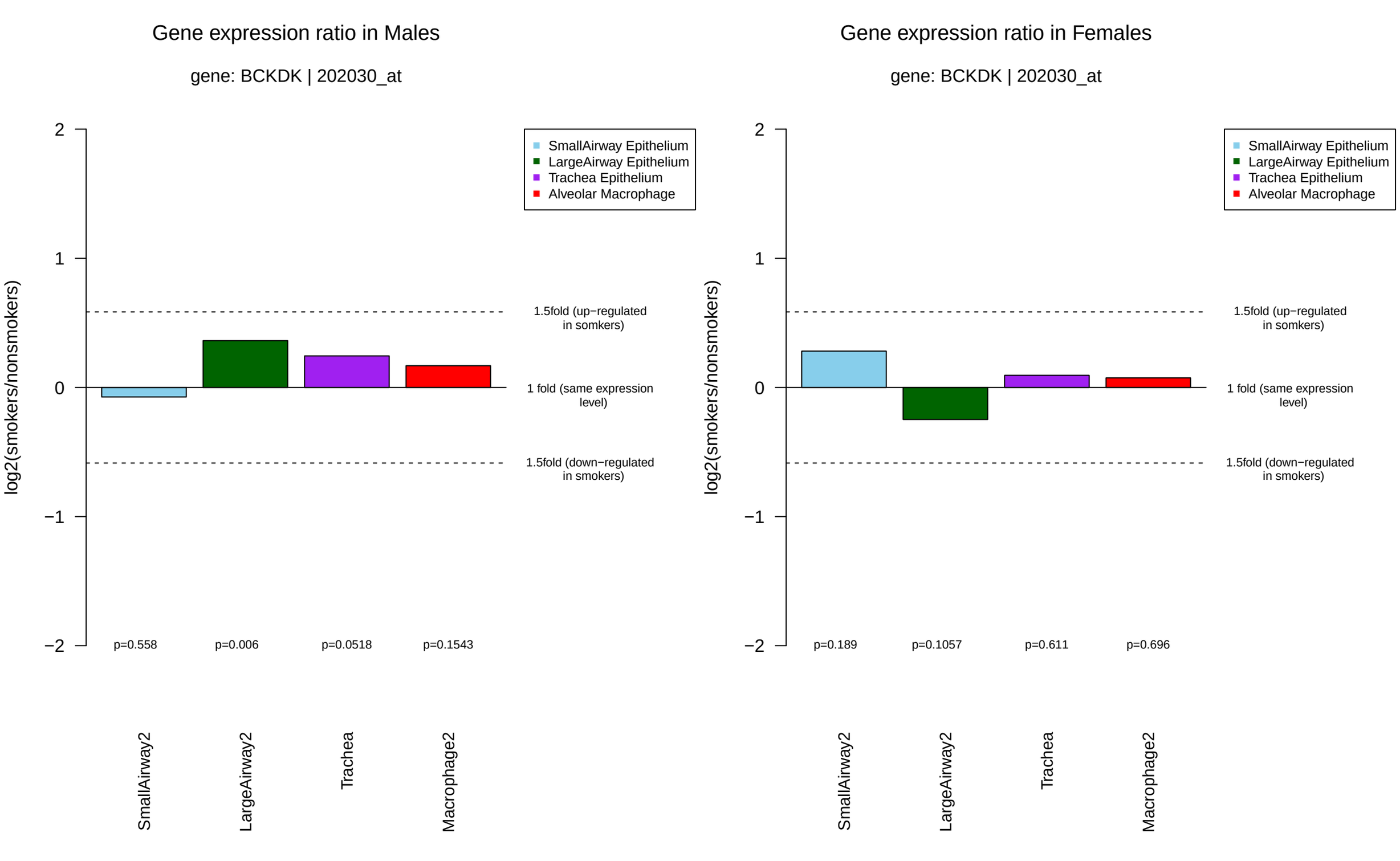
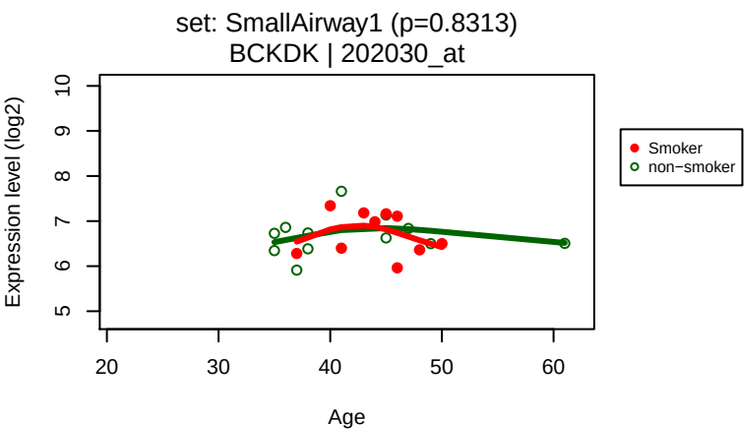
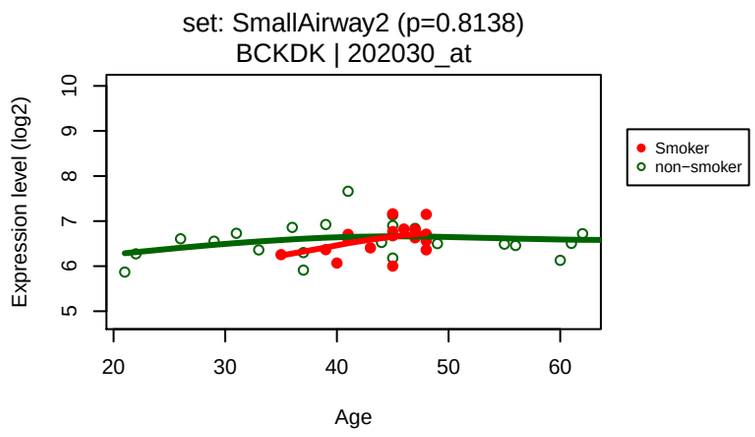
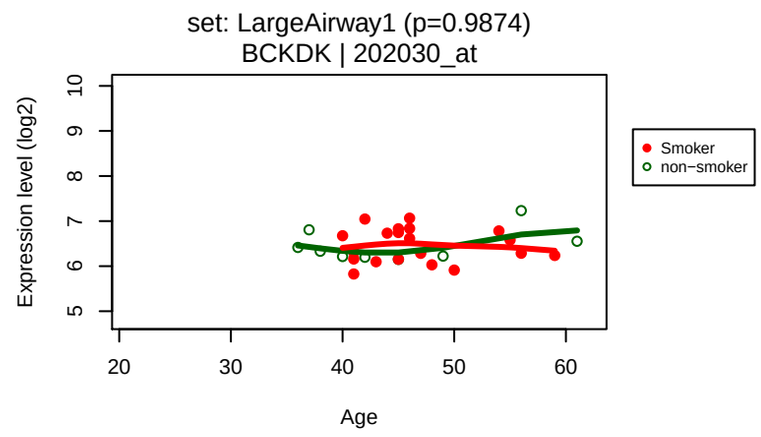

Gene expression of smokers and nonsmokers
gene: BCKDK | 202030_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8313
p=0.8138
p=0.5883
p=0.9874
p=0.1473
p=0.02
p=0.7778
p=0.1392
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

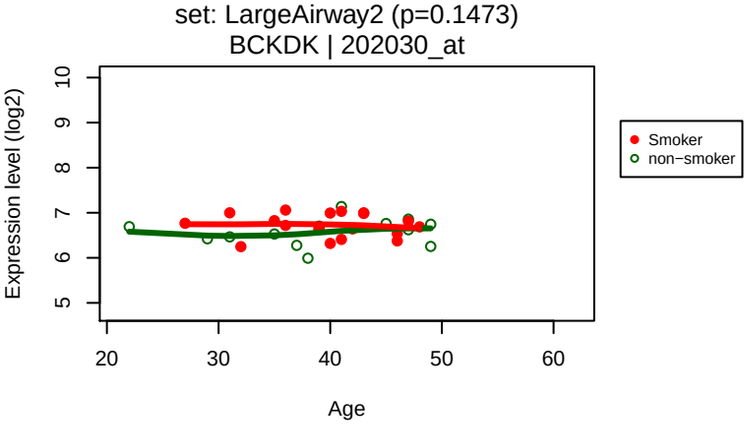

Gene expression ratio (smokers/nonsmokers)
gene: BCKDK | 202030_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8313
p=0.8138
p=0.5883
p=0.9874
p=0.1473
p=0.02
p=0.7778
p=0.1392
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

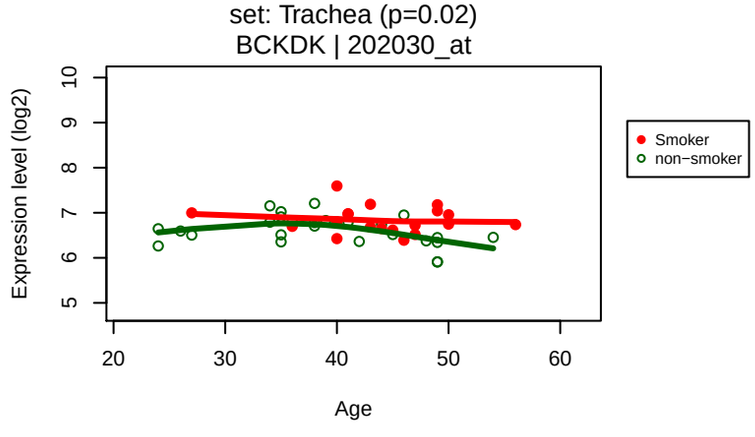

Gene expression ratio in Males
gene: BCKDK | 202030_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.558
p=0.006
p=0.0518
p=0.1543
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BCKDK | 202030_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.189
p=0.1057
p=0.611
p=0.696
SmallAirway2
LargeAirway2
Trachea
Macrophage2

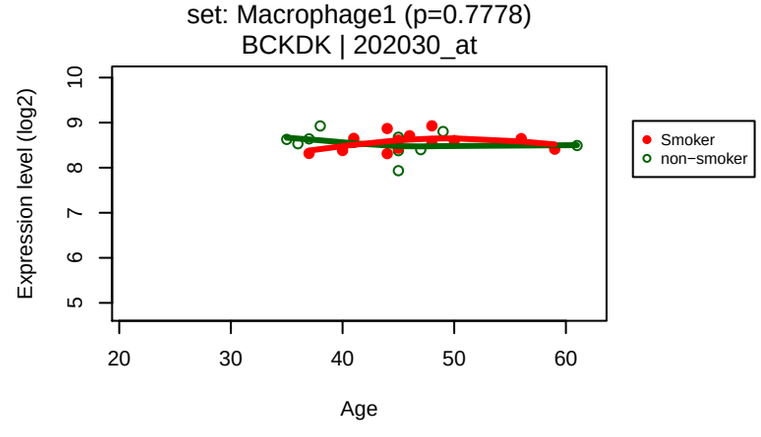

set: SmallAirway1 (p=0.8313)
BCKDK | 202030_at
Expression level (log2)
Age
Smoker
non−smoker

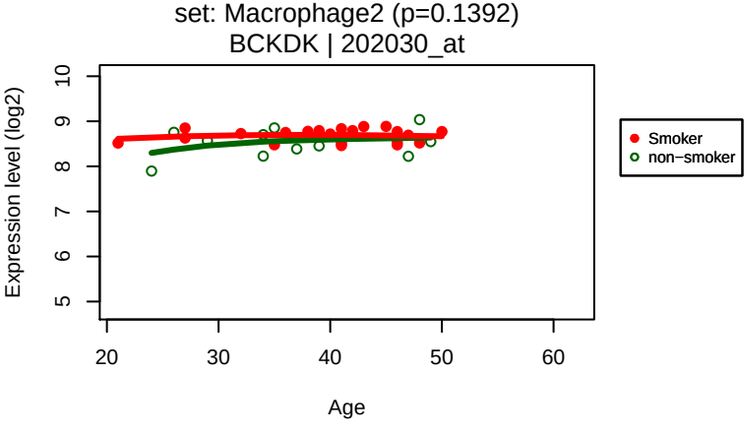

set: SmallAirway2 (p=0.8138)
BCKDK | 202030_at
Expression level (log2)
Age
Smoker
non−smoker

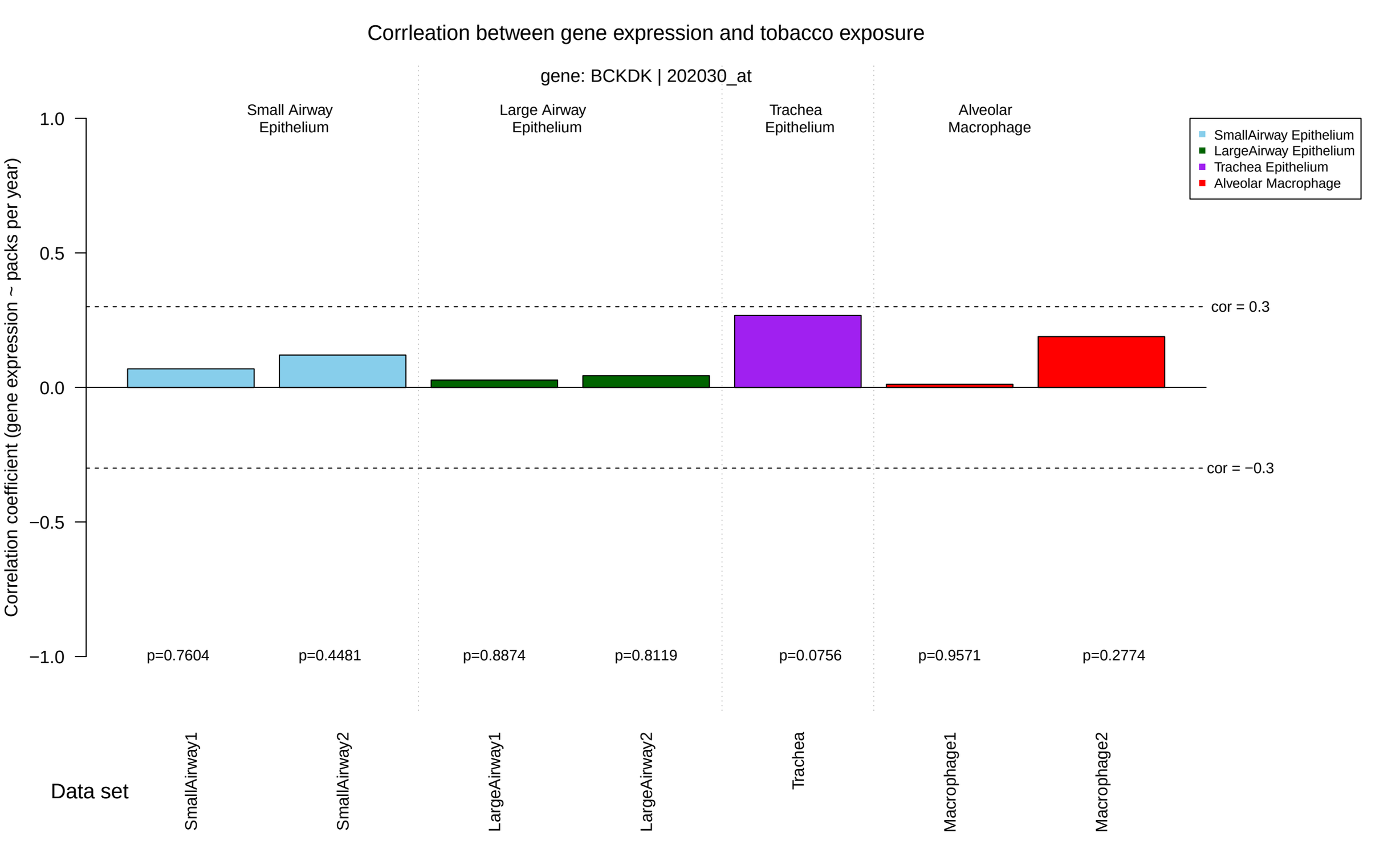

set: LargeAirway1 (p=0.9874)
BCKDK | 202030_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1473)
BCKDK | 202030_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.02)
BCKDK | 202030_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7778)
BCKDK | 202030_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1392)
BCKDK | 202030_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BCKDK | 202030_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7604
p=0.4481
p=0.8874
p=0.8119
p=0.0756
p=0.9571
p=0.2774
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2